Biochemistry Cume

Understanding metabolic pathways is essential in fields like medicine and biotechnology. It helps explain diseases caused by metabolic defects and guides the development of therapeutic interventions.

Trace the fate of carbon atom number 1 (C1 labeled with ^{14}C) in a glucose molecule as it undergoes glycolysis and is eventually converted to alanine (Ala) through the process of gluconeogenesis. Provide a step-by-step account of the relevant reactions and intermediates involved in both pathways.

Keep in mind that this is a detailed question that requires a comprehensive understanding of glycolysis and gluconeogenesis, as well as the conversion of glucose to alanine. Label the location of ^{14}C in the intermediates (9pts each).